Data Analyst Capstone Project

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EXECUTIVE SUMMARY



key findings

- Programming Languages: JavaScript is the most-used language, followed by HTML/CSS and SQL.
- Databases: PostgreSQL leads current databases, followed by MYSQL and SQLite.
- Platforms: AWS is the leading cloud/platform technology in the current-usage analysis.
- Web Frameworks: Node.js and React are the leading web technologies in the dashboard.
- Overall: The results highlight strong demand for web development, database, and cloud technologies.

INTRODUCTION



Purpose: Analyze current and future technology trends among developers and technology professionals.

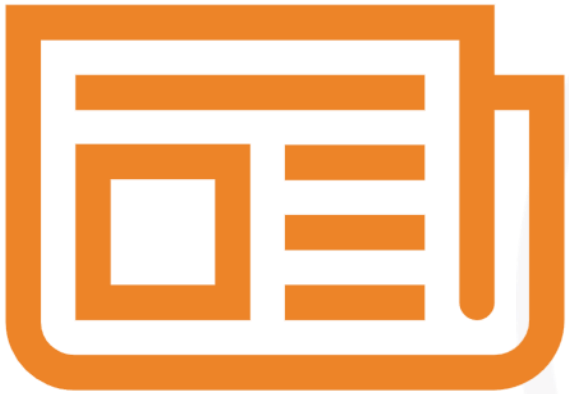
Focus: Examine programming languages, databases, platforms, and web frameworks.

Target Audience: Developers, data professionals, employers, and individuals planning technology careers.

Value: Identify technologies that are widely used today and those expected to grow in demand.

Outcome: Provide data-driven insights to support technology skills and career planning.

METHODOLOGY



Data sources, Data collection & Data wrangling

Developer survey data was used for the analysis.

Data was cleaned and prepared before visualization.

Multiple-response technology fields were separated and analyzed.

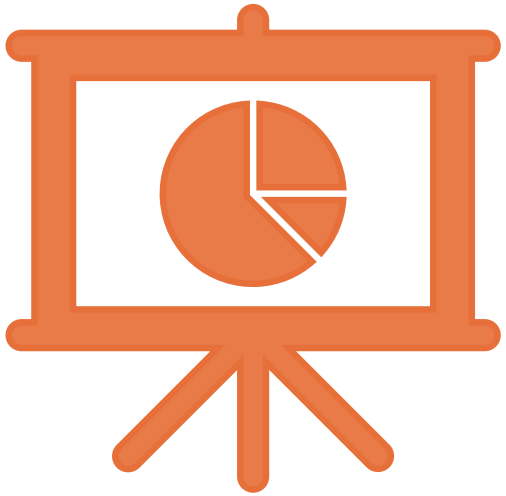
Technologies were ranked by respondents count.

- Current technology usage was analyzed.
- Future technology preferences were analyzed.

Demographic characteristics were analyzed by age, country, and education.

IBM COgnos Analytics was used to create the dashboards and visualizations.

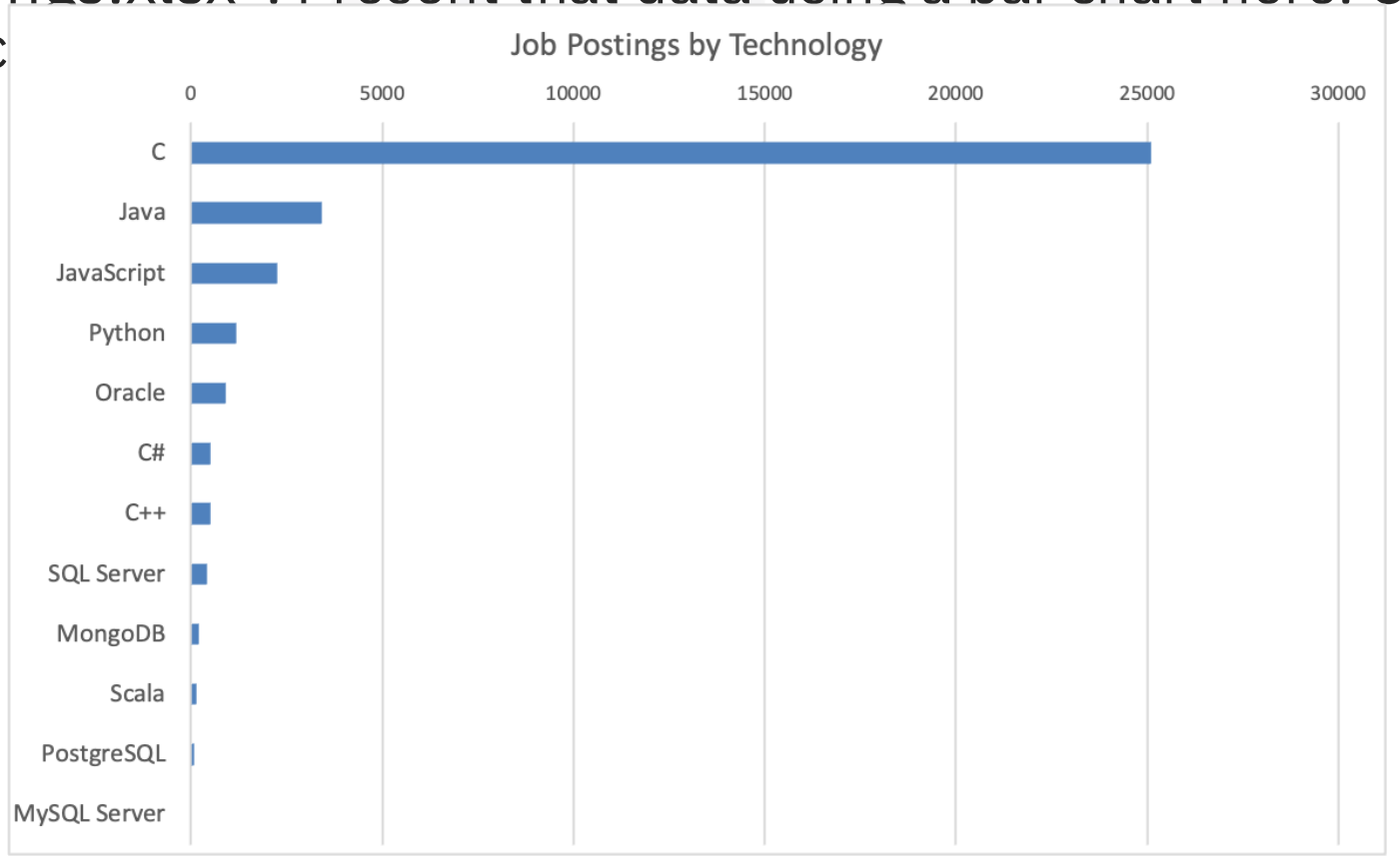
Results



Results from the analysis of current technology usage, fyture technology trends, and respondents demographics are presented in the following slides.

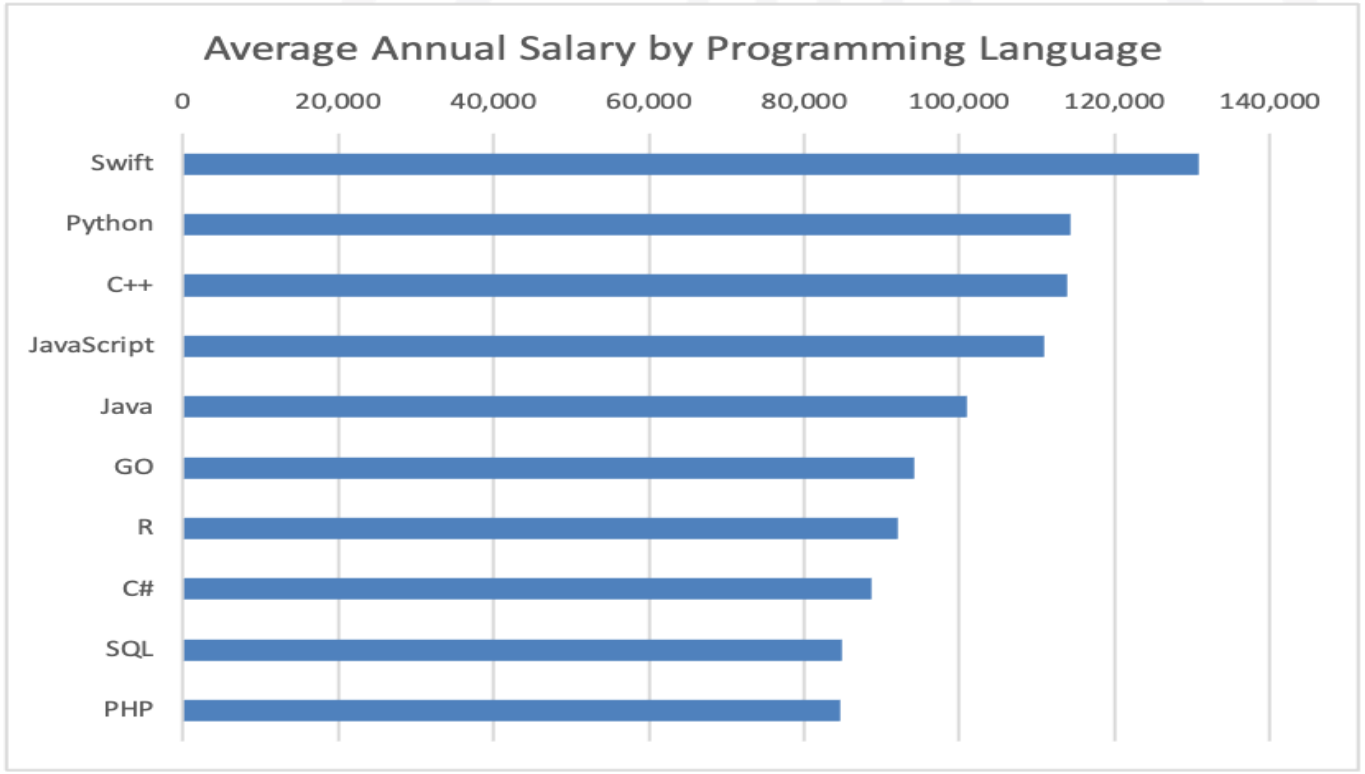
JOB POSTINGS

In Module 1 you have collected the job posting data using Job API in a file named "job-postings.xlsx". Present that data using a bar chart here. Order the bar chart in the desc



POPULAR LANGUAGES

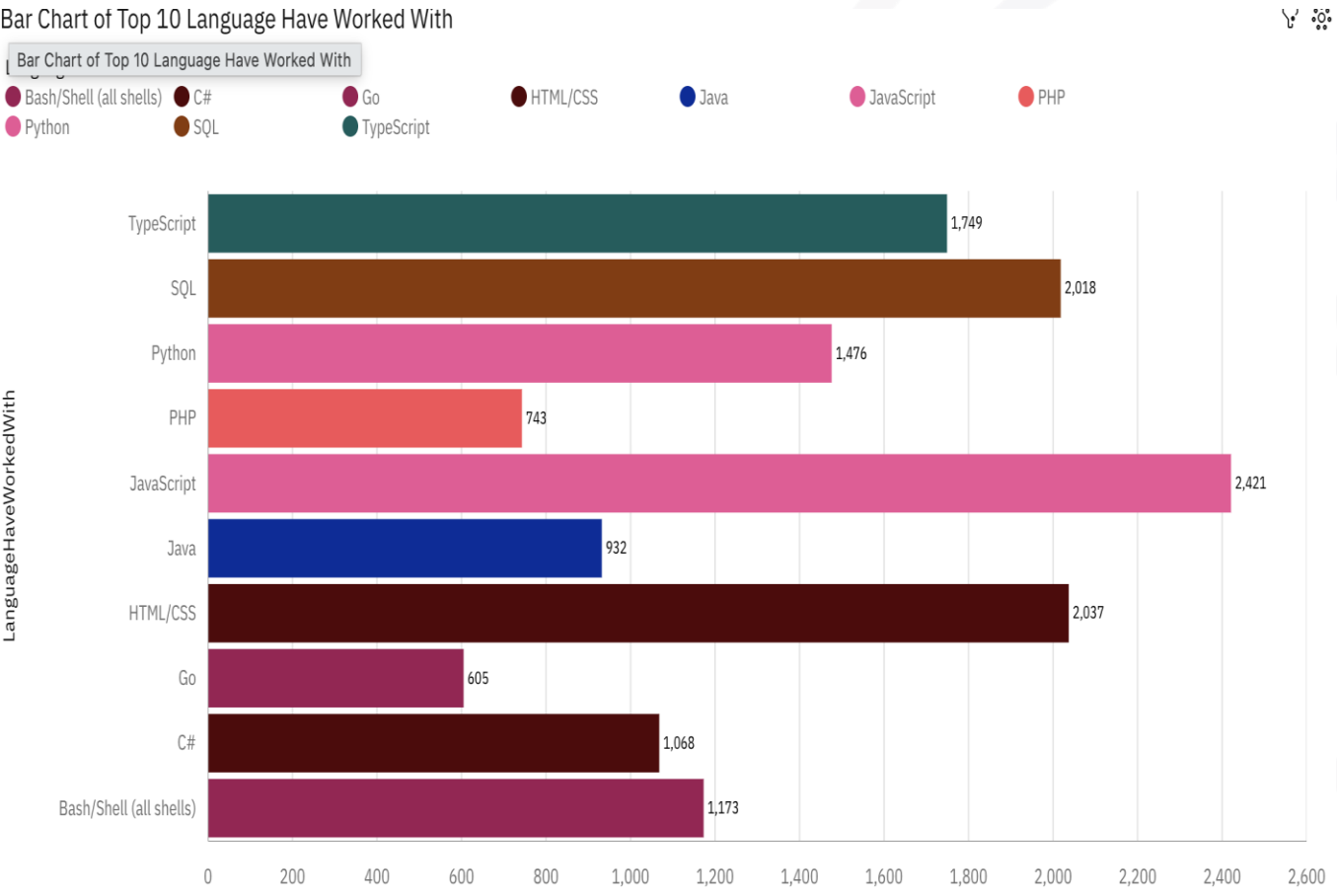
In Module 1 you have collected the job postings data using web scraping in a file named “popular-languages.csv”. Present that data using a bar chart here. Order the bar chart in the descending order of salary.



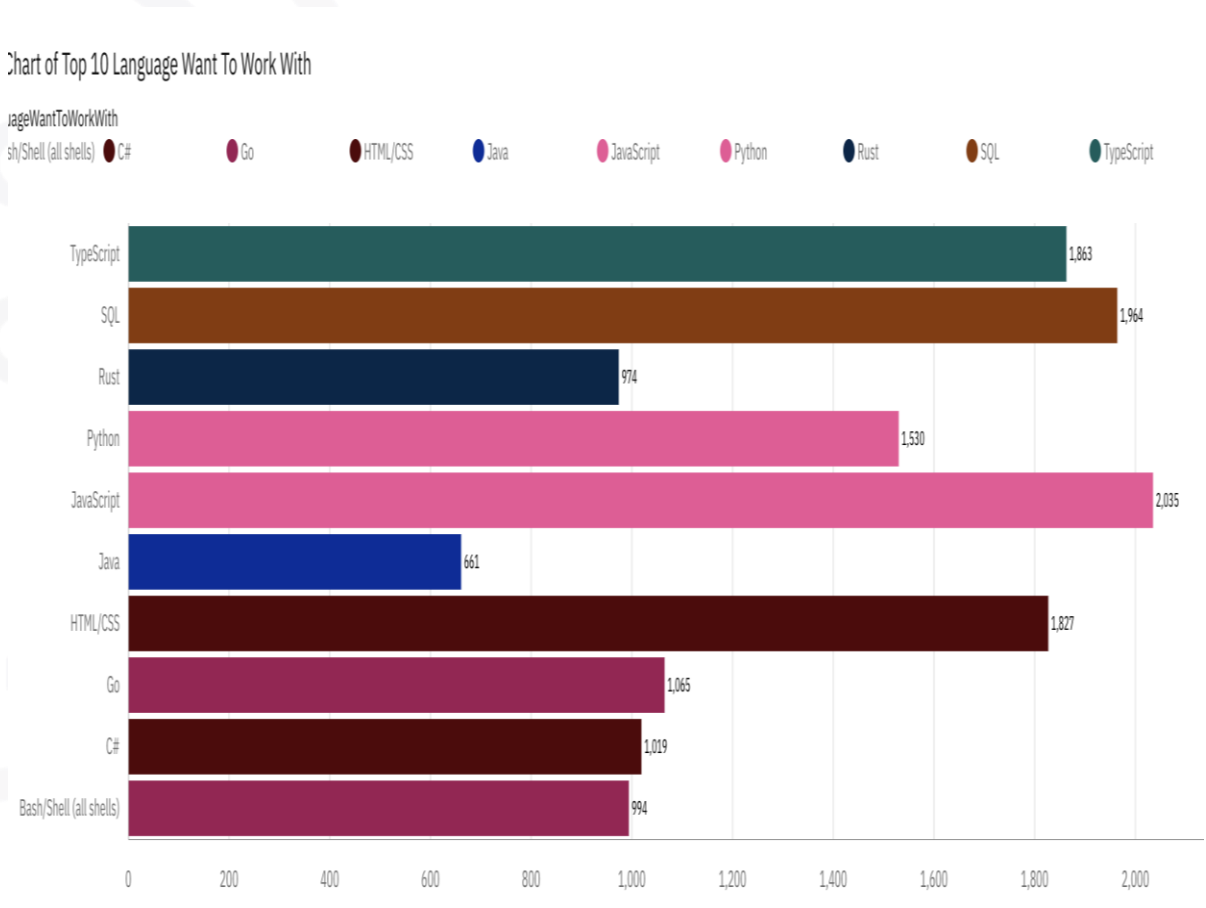
PROGRAMMING LANGUAGE TRENDS

JavaScript remains highly popular, while SQL,HTML/CSS, and TypeScript also show strong demand. Python remains an important language for future use.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript has the highest current usage.
- SQL and HTML/CSS also have high current usage.
- JavaScript, SQL, and TypeScript show strong future interest.

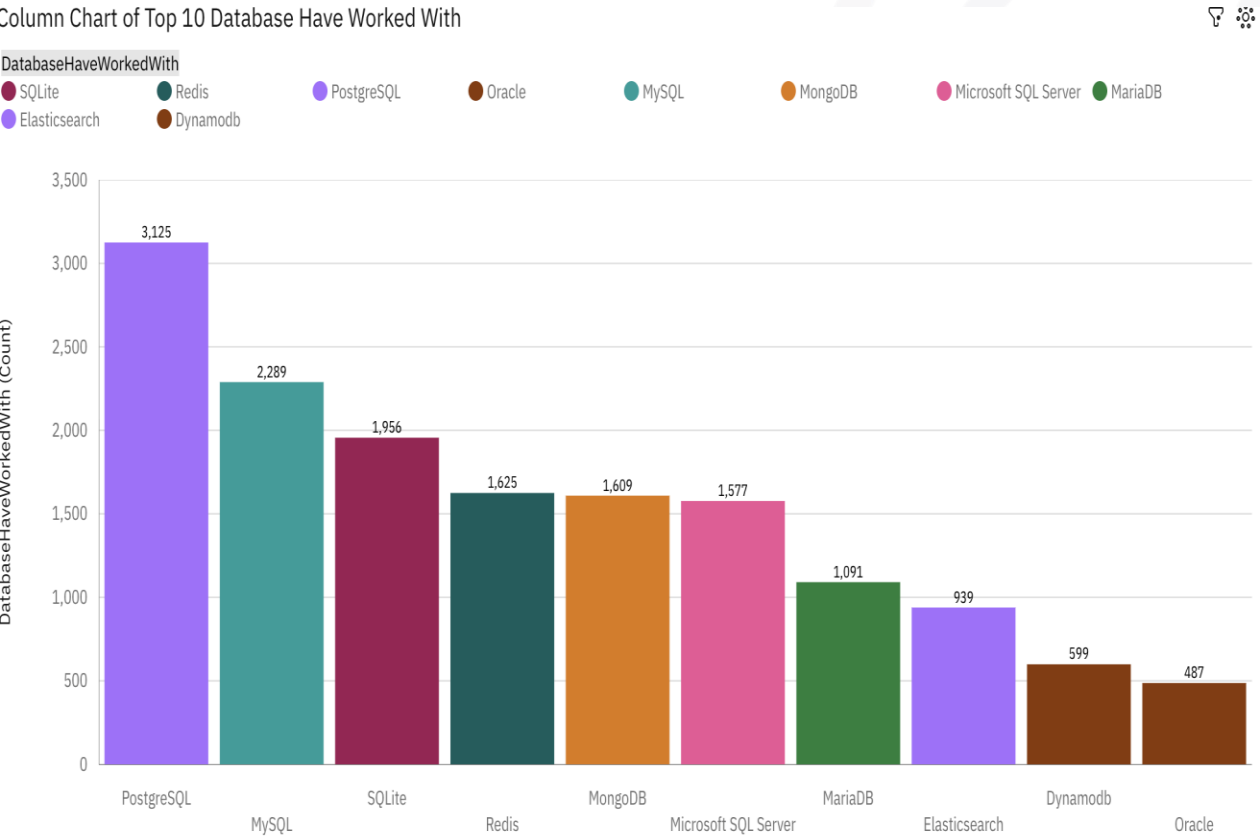
Implications

- JavaScript remains an important skill for developers.
- SQL and web technologies remain valuable.
- TypeScript shows strong interest for future development.

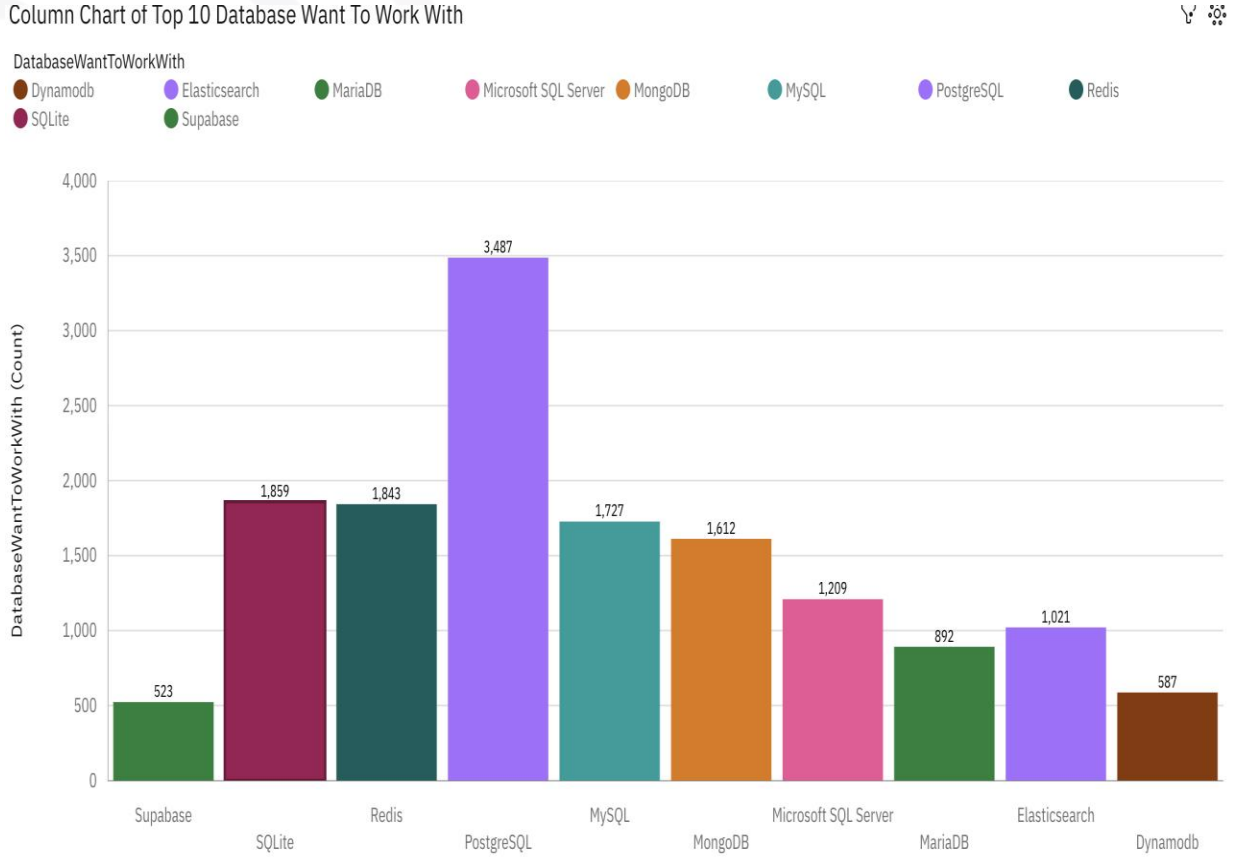
DATABASE TRENDS

PostgreSQL leads both current database usage and future interest. MySQL, and SQLite also remain widely used.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL has the highest current usage.
- MySQL and SQLite are also widely used.
- PostgreSQL leads future database interest.

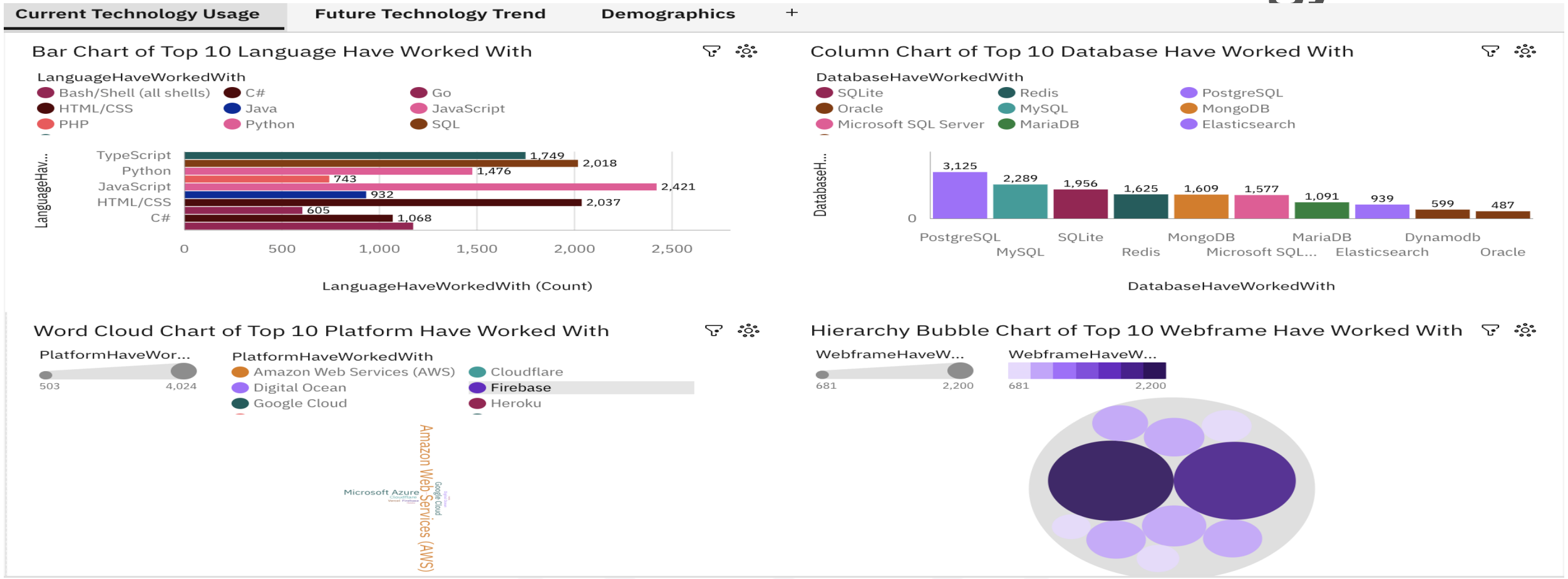
Implications

- PostgreSQL remains a key database skill.
- MySQL and SQLite remain valuable.
- PostgreSQL shows strong demand for future development.

DASHBOARD

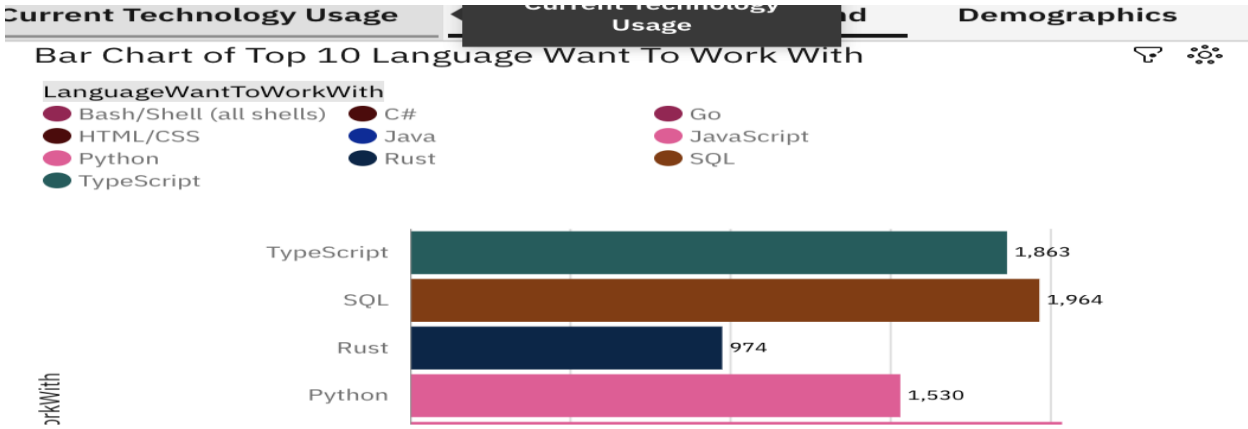


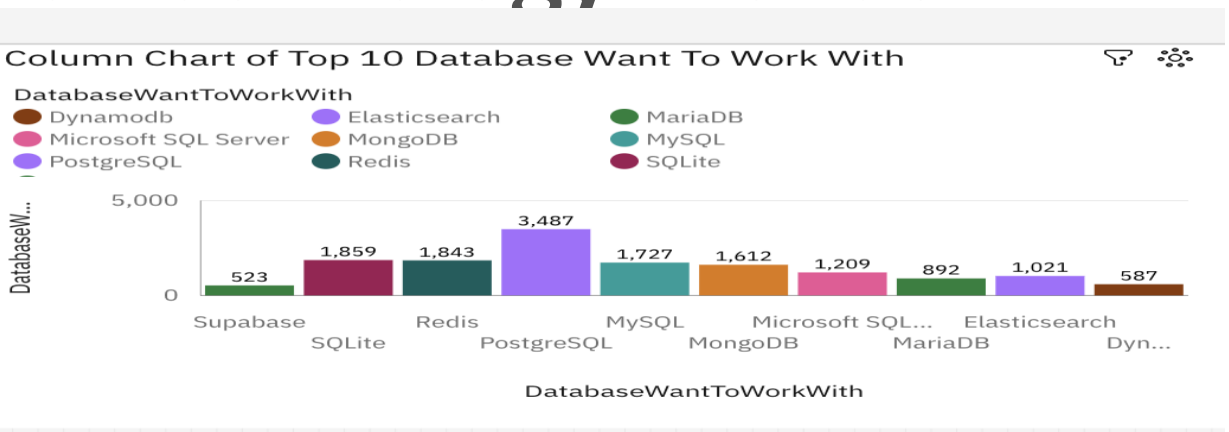
DASHBOARD TAB 1: Current Technology

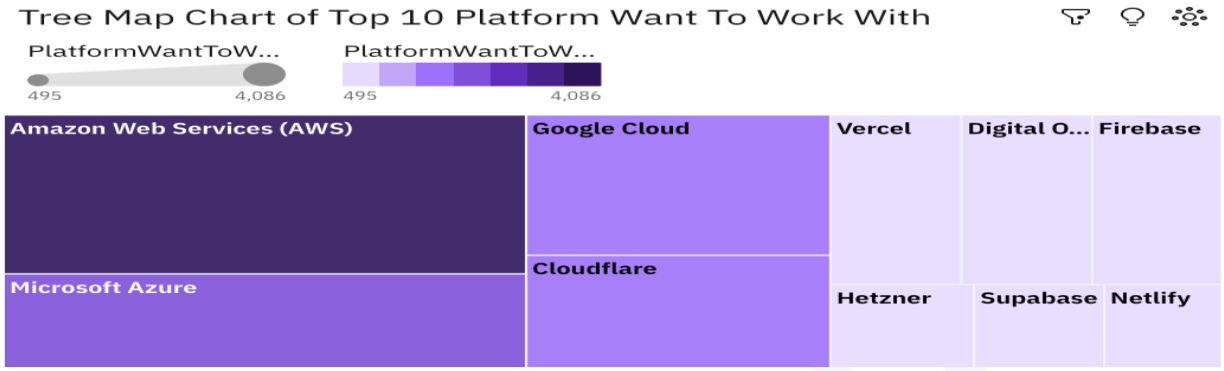


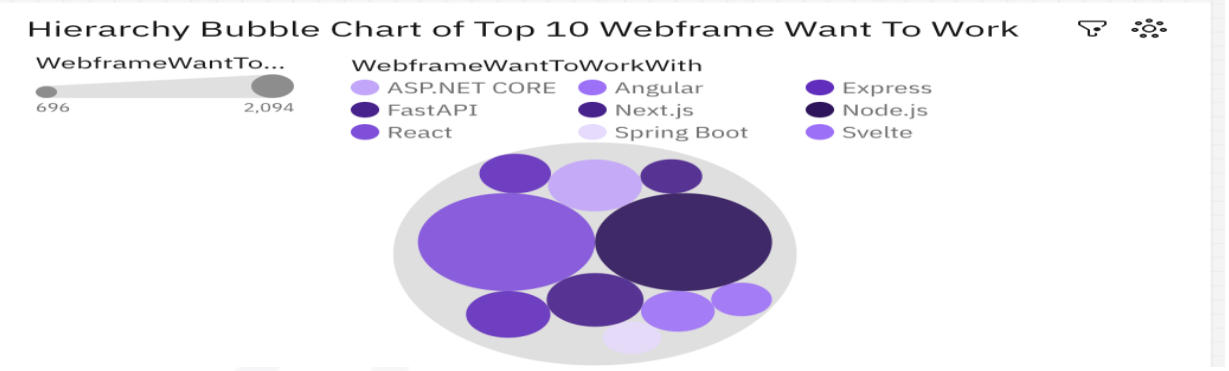
- Dashboard Insight: JavaScript, HTML/CSS, and SQL are among the most commonly used technologies, while PostgreSQL leads current database usage.

DASHBOARD TAB 2: Future Technology Trends



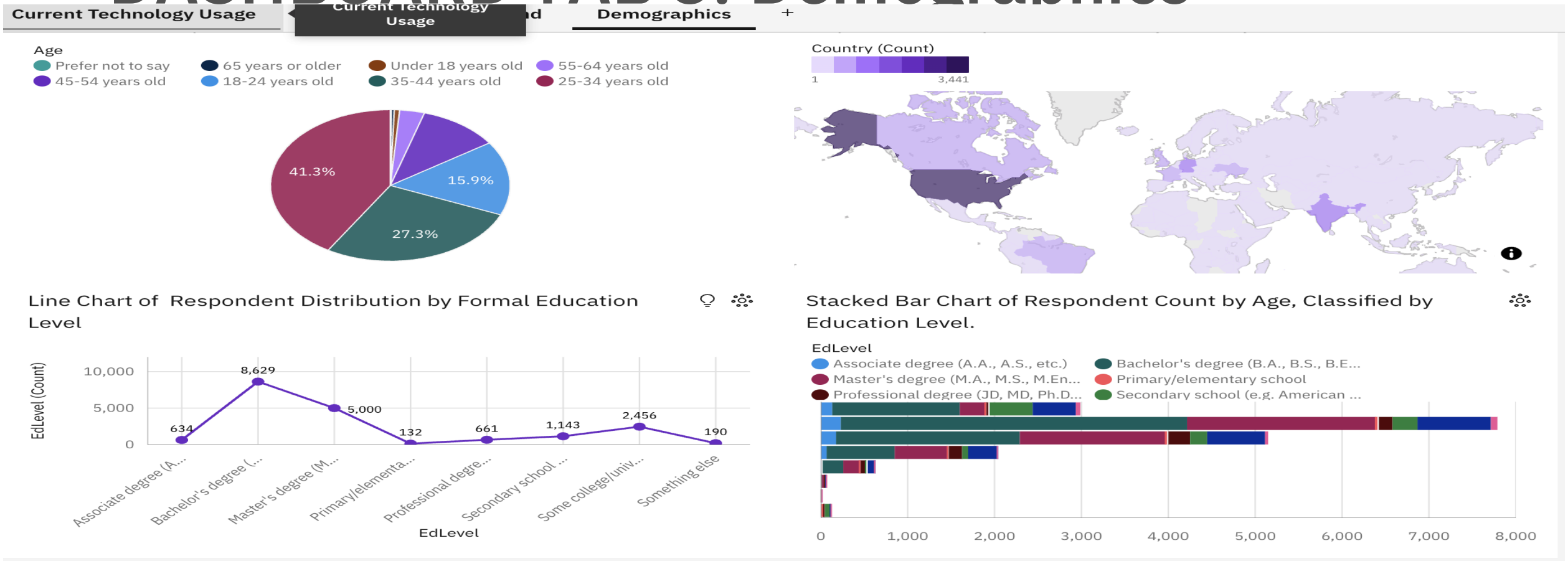






- Dashboard Insight: SQL, TypeScript, and Python show strong future interest, while PostgreSQL leads future database demand.

DASHBOARD TAB 3: Demographics



Dashboard Insight: Most respondents are between ages 25-44, bachelor's degrees are the most common education level, and the United States has the highest respondent count.

DISCUSSION



- JavaScript, HTML/CSS, and SQL are widely used, showing the continued importance of core web and database skills.
- SQL, TypeScript, and Python show strong future interest, while PostgreSQL leads database interest.
- Most respondents are ages 25-44,with a bachelor's degree as the most common education level and the U.S. having the highest respondent count.

OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript, HTML/CSS, and SQL are widely used.
- SQL, TypeScript, and Python show strong future interest.
- PostgreSQL leads both current usage and future database interest.

Implications

- Core web and SQL skills remain important.
- Python and TypeScript are valuable skills for future development.
- PostgreSQL is an important database skill to develop.

CONCLUSION



- JavaScript, HTML/CSS, and SQL remain widely used technologies.
- SQL, TypeScript, and Python show strong future interest.
- PostgreSQL leads both current use and future database interest.
- The results highlight the importance of continuously developing programming and database skills.

APPENDIX

